SMPS AND UPS

Easy-to-Learn Computer Hardware Assignment

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Class: _____ **Date:** _____

Learning goal: Understand SMPS and UPS, recognize common power connectors, and learn the safe order for connecting a gaming PC.

Quick memory tip

SMPS = correct power. It converts AC electricity into the DC voltages computer parts need.
UPS = backup power. It gives temporary power when the main electricity supply fails.

1. Devices that use SMPS and UPS

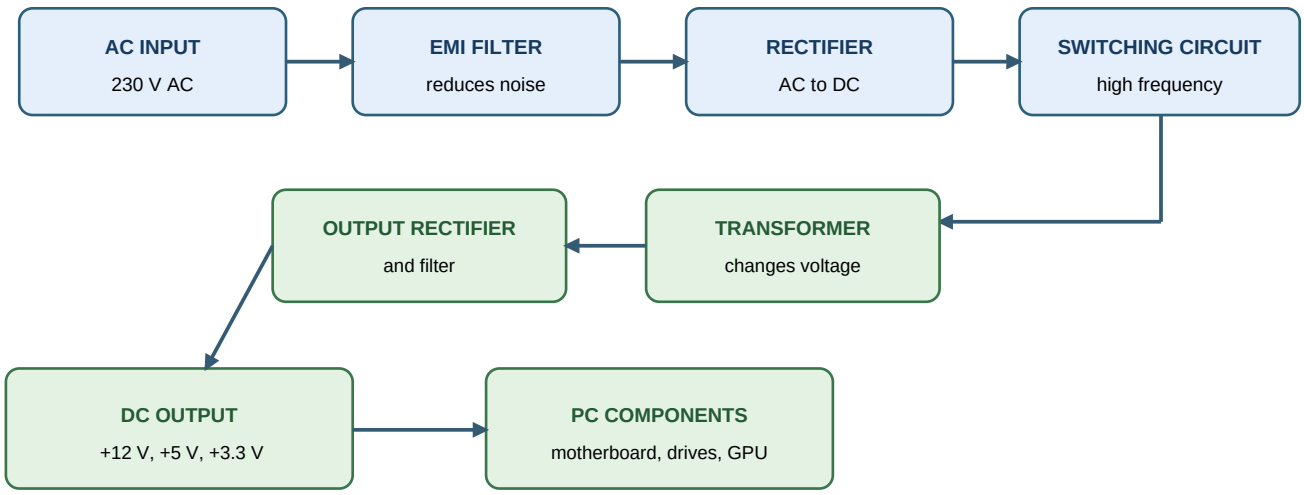
SMPS means **Switched Mode Power Supply**. UPS means **Uninterruptible Power Supply**.

No.	Device	Type	Why it needs it
1	Desktop computer	SMPS	Converts AC electricity into DC power for computer parts.
2	Laptop charger	SMPS	Changes electricity to the voltage needed to charge the laptop.
3	Printer	SMPS	Supplies suitable power to its electronic circuits.
4	Desktop PC	UPS	Provides temporary backup power during a power cut.
5	Wi-Fi router	UPS	Can keep the router working during a power cut if the UPS has enough capacity.

Remember: SMPS supplies the correct power; UPS provides backup power.

2. Diagram of a desktop PC SMPS

Follow the arrows to understand how electricity moves through the SMPS.



SMPS converts AC electricity into stable DC power.

Never open the SMPS case; internal capacitors can retain dangerous charge.

Easy explanation: AC electricity enters the SMPS. The rectifier changes it into DC, the switching circuit and transformer help change the voltage, and the output rectifier/filter produces stable DC outputs for PC components.

3. Four common power connectors

The simple drawings below show the general shape of each connector.

ATX 24-pin

Devices it powers: Powers the motherboard.

Safety tip: Turn off and unplug the PC; press the connector latch before removing it.

SATA power

Devices it powers: Powers SATA hard drives (HDDs) and SATA solid-state drives (SSDs).

Safety tip: Match the L-shaped connector and never force it.

Molex

Devices it powers: Powers some older drives, fans and accessories.

Safety tip: Hold the plastic plug, not the wires, when disconnecting.

PCIe 6/8-pin

Devices it powers: Provides extra power to compatible graphics cards (GPUs).

Safety tip: Use the correct GPU cable; do not substitute a CPU/EPS cable or force the plug.

4. Connecting SMPS cables in a new gaming PC

Always keep the PC switched off and unplugged while connecting cables. Check the PSU, motherboard and graphics card manuals because requirements differ by model.

1. Switch off the power

Turn off the PSU, unplug the power cord and work on a stable surface. Never open the PSU case.

2. Connect the motherboard

Use the large **24-pin ATX cable**. Align it with the motherboard socket and push gently until the latch clicks.

3. Connect CPU power

Use the **4-pin or 8-pin CPU/EPS cable** near the CPU socket. Do not confuse it with the PCIe GPU cable.

4. Connect the graphics card

If required, use the correct **PCIe 6-pin or 8-pin cable**, or the exact GPU power connector specified by the manufacturer.

5. Connect storage devices

For a SATA HDD or SATA SSD, connect a **SATA power cable** from the PSU and a separate SATA data cable to the motherboard. An M.2 NVMe SSD usually needs no separate power cable.

6. Check the cables

Make sure plugs are fully seated and cables are clear of fans. Use only modular PSU cables approved for that exact PSU model.

7. Start the PC

Close the case, reconnect the power cord, switch on the PSU and start the PC. If you notice sparks, burning smell or unusual noise, switch off and unplug immediately.

Quick revision

Connector / device	Main purpose
24-pin ATX	Main motherboard power
CPU/EPS 4/8-pin	CPU power near the CPU socket
PCIe 6/8-pin	Extra power for a graphics card
SATA power	Power for SATA storage drives
SMPS	Supplies the correct DC voltages
UPS	Provides temporary backup power

Conclusion: SMPS provides the correct DC power to computer parts, while UPS provides temporary backup during a power cut. Correct connectors and safe handling help protect both the computer and the user.